ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MATHEMATICAL LITERACY

PRACTICE EXAM

MARKS: 100 **TIME: 3 hours**

NC(V) fundamental subject, Level 2. Finance (30), measurement (25), maps, plans and scale (15), data handling (20), probability (10). All people, prices and data are fictional and were created for this examination.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number the answers according to the numbering system used in this question paper.
3. Show ALL calculations. Answers only will not necessarily earn full marks.
4. An approved calculator (non-programmable and non-graphical) may be used.
5. Round off answers to TWO decimal places, unless stated otherwise. Write units with all answers.
6. Write neatly and legibly.

QUESTION 1: FINANCE

1.1 The Dlamini family's monthly income and expenses are shown below.

Income	Amount (R)	Expenses	Amount (R)
Salary (Mrs Dlamini)	8 500	Rent	3 200
Child support grant	560	Groceries	2 450
		Transport	1 320
		Electricity	600
		Airtime and data	250
		School fees	400

1.1.1 Calculate the total monthly income. (2)

1.1.2 Calculate the total monthly expenses. (2)

1.1.3 Does the family have a surplus or a deficit? Calculate the amount. (2)

1.1.4 What percentage of the income is spent on rent? (2)

1.1.5 Classify rent and groceries as fixed or variable expenses. (2)

1.1.6 Suggest TWO ways in which the family can reduce its expenses. (2)

1.2 Prepaid electricity costs R2,85 per kWh. How many kWh will Mrs Dlamini receive for R300? (2)

1.3 Sipho's monthly gross salary is R12 400. His deductions are: UIF 1% of gross salary, PAYE R1 150 and medical aid R650.

1.3.1 Calculate his UIF contribution. (2)

1.3.2 Calculate his total deductions. (2)

1.3.3 Calculate his net salary. (2)

1.3.4 Explain the difference between gross and net salary. (2)

1.4 VAT in South Africa is 15%.

1.4.1 A toolbox costs R480 excluding VAT. Calculate the price including VAT. (2)

1.4.2 A drill costs R920 including VAT. Calculate the price excluding VAT. (2)

1.5 A shop offers a 25% discount on a R1 200 jacket. Calculate the sale price. (2)

1.6 Zanele borrows R5 000 from a micro-lender at 18% simple interest per annum for 18 months. Calculate the total amount she must repay. (2)

[30]

QUESTION 2: MEASUREMENT

2.1 Convert: (a) 2,5 kg to g (b) 750 mℓ to ℓ (c) 1,8 m to cm (d) 3 hours 45 minutes to minutes (4)

2.2 A room is 4 m long, 3,5 m wide and 2,6 m high. The four walls must be painted. The door (1,8 m²) and window (1,5 m²) will not be painted.

2.2.1 Calculate the perimeter of the floor. (2)

2.2.2 Calculate the area of the walls to be painted. (3)

2.2.3 One litre of paint covers 8 m². Two coats are needed. Calculate the number of litres needed. (3)

2.2.4 Paint is sold in 5 ℓ tins at R420 each. How many tins must be bought, and what will they cost? (3)

2.3 A bus leaves the college at 06:45 and arrives in town at 08:10. How long is the trip? (2)

2.4 A recipe for 4 people uses 300 g of flour. How much flour is needed for 10 people? (2)

2.5 Use the formula °F = (9 ÷ 5 × °C) + 32 to convert 30 °C to degrees Fahrenheit. (2)

2.6 A water tank holds 2 500 ℓ. A family uses 180 ℓ per day. For how many full days will a full tank last? (2)

2.7 A rectangular garden is 12 m by 8 m. Calculate how many metres of fencing are needed and the cost if fencing costs R65 per metre. (2)

[25]

QUESTION 3: MAPS, PLANS AND SCALE

3.1 On a map with a scale of 1 : 50 000, the distance between two villages is 7,4 cm.

3.1.1 Calculate the real distance in kilometres. (3)

3.1.2 How long (in minutes) will it take to walk this distance at 5 km/h? (2)

3.1.3 Explain the meaning of the scale 1 : 50 000. (2)

3.2 On a floor plan with a scale of 1 : 100, a bedroom measures 4,5 cm by 3 cm.

3.2.1 Calculate the real length and width of the bedroom in metres. (2)

3.2.2 Calculate the real floor area of the bedroom. (2)

3.3 Give TWO advantages of a bar scale over a number scale. (2)

3.4 Name the compass direction that is exactly opposite north-east. (2)

[15]

QUESTION 4: DATA HANDLING

4.1 Thirty students were asked how many hours they spend on their phones per day. The results are:

Hours per day	**1**	**2**	**3**	**4**	**5**	**6**
Number of students	2	5	8	7	5	3

4.1.1 Is the data discrete or continuous? Give a reason. (2)

4.1.2 What is the modal number of hours? (1)

4.1.3 Calculate the mean number of hours per student. (4)

4.1.4 Determine the median. (2)

4.1.5 What percentage of the students spend 5 hours or more on their phones? (2)

4.1.6 Draw a bar graph of the data. (4)

4.2 A company advertises: "9 out of 10 users love our product!" Give TWO questions you would ask to check whether this claim is reliable. (2)

4.3 Give ONE advantage of a questionnaire and ONE disadvantage of using only your friends as a sample. (3)

[20]

QUESTION 5: PROBABILITY

5.1 A bag contains 4 red, 6 blue and 10 white tokens. One token is taken at random. Calculate, as a fraction in simplest form, the probability that it is: (a) blue (b) not white (c) green (4)

5.2 A weather forecast says there is a 70% chance of rain tomorrow. Explain what this means. (2)

5.3 A coin is tossed and a dice is rolled. List all the possible outcomes and calculate the probability of getting heads and an even number. (4)

[10]

TOTAL: 100

END OF QUESTION PAPER